

Dynamic Sparse Training in Continual Learning

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Introduction

Motivation

Is DST explorative in nature?

- Existing research suggest that dynamic sparse training has the ability to compete with dense neural networks, as it may be less prone to overfitting and instability. So, we wish to investigate if, when faced with non-stationary dataset, will DST be able to learn without suffering from instability? In generally how plastic is DST when faced with new data? In comparison to sparse training methods such as GMP and Static, DST seems more explorative; but will this allow it to adapt to new data over time? Is sparsity alone enough to deal with non-stationary data? This leads to our next big question, how does all sparsely trained model fare against non-stationary data; does DST's unique behavior bring an edge when dealing with new data.
- State-of-the-art Dynamic Sparse Training (DST) methods learn *unstructured* Sparse Neural Networks (SNNs) with 85–95% fewer weights than dense models, while maintaining similar generalization.
- Sparse training methods employ sparsity *both during training and inference*, unlike pruning and other methods [?] that only exploit sparsity at inference time.
- While unstructured SNNs enable a large reduction in Floating Point Operations (FLOPs) in theory, realizing these speedups on hardware is challenging.
- Structured sparse pruning often results in worse generalization than unstructured SNNs.

Method

- We propose a novel sparse-to-sparse DST method, Structured RigL (SRigL), based on RigL [?]. SRigL learns a SNN with constant fan-in structured sparsity while maintaining generalization comparable with RigL up to a high sparsity level (99%) for a variety of network architectures. This structure is a particular case of “N:M sparsity” [?].
- We show that at sparsity levels >90% RigL ablates whole neurons. By allowing ablation in SRigL, we match the generalization performance of RigL even in this high-sparsity regime.
- We motivate our choice of structure with a theoretical analysis of SNN output norm variance and find that the constant fan-in constraint does not have a negative effect.
- We demonstrate that our constant fan-in sparsity enables a compact representation that is not only parameter- and memory-efficient, but also amenable to real-world acceleration.

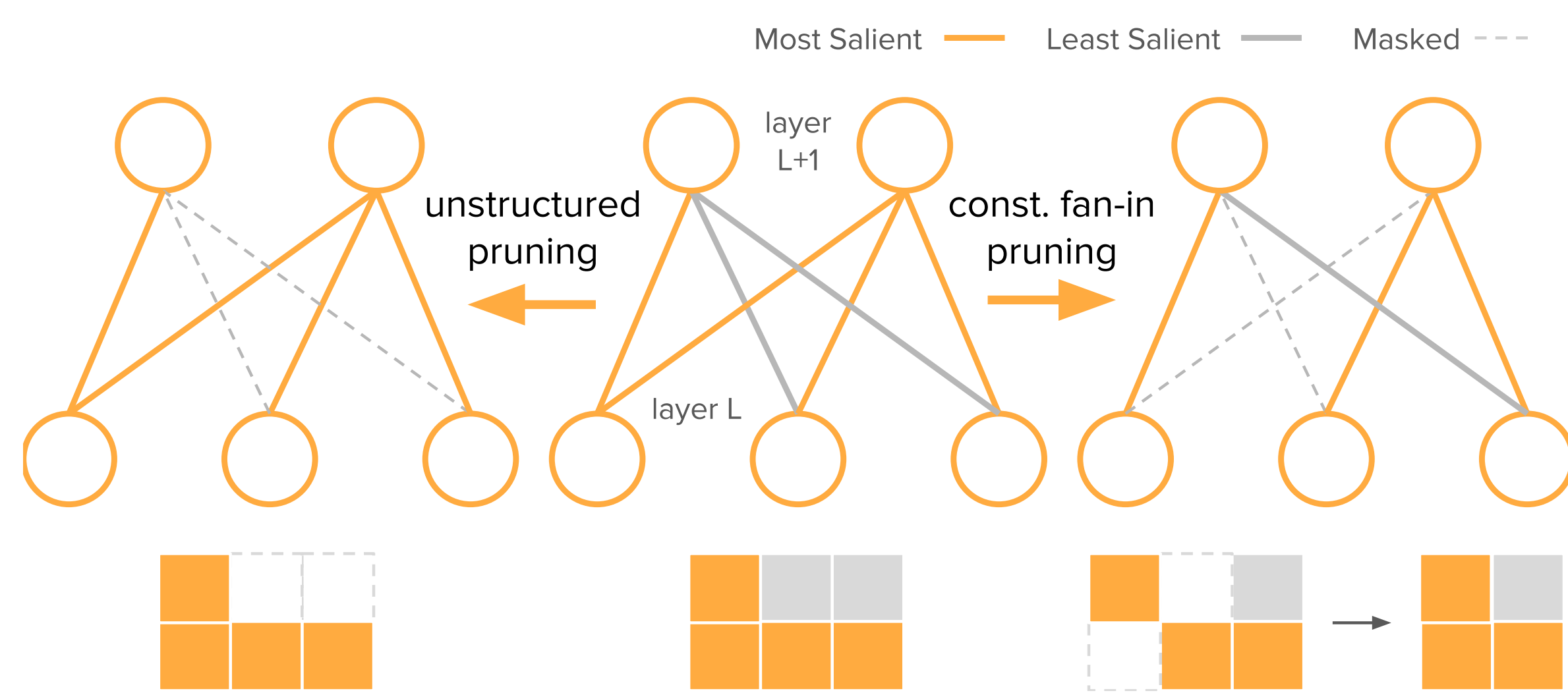
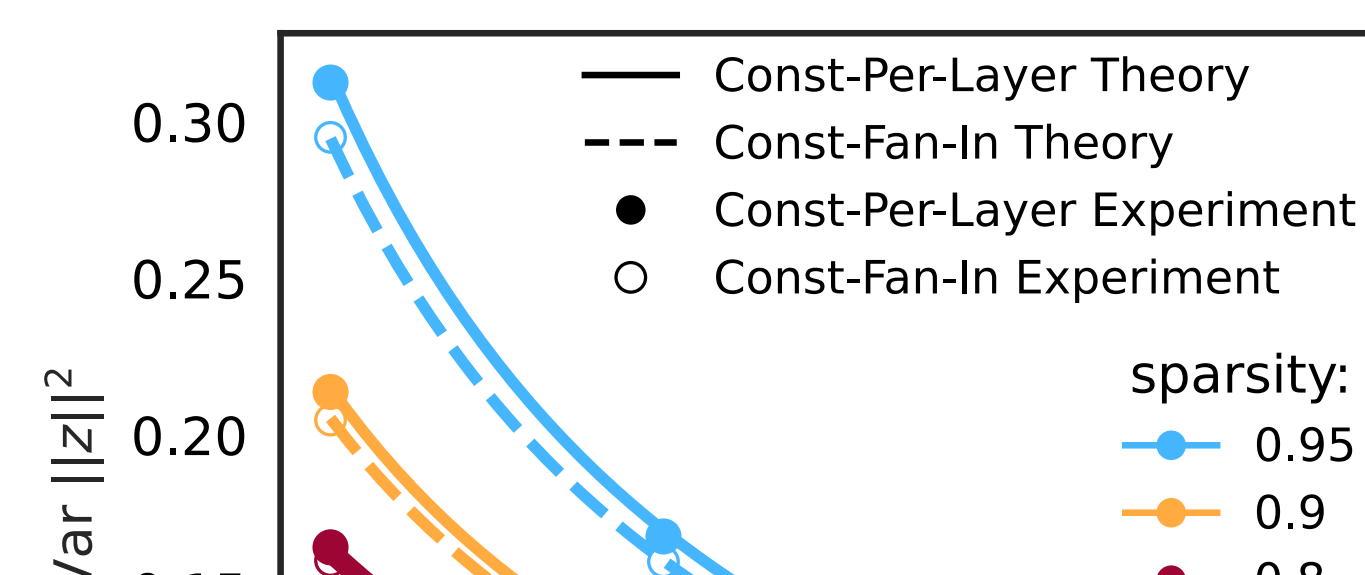


Figure 1. Constant fan-in pruning v.s. unstructured pruning.

Evaluation

- Smaller output variance is indicative of training stability
- We derive exact expressions for the variance of the layer output norm at initialization
- Constant fan-in consistently yields smaller variance than other sparsity types



Results

Table 1. Top-1 ImageNet test accuracy of ResNet-50 trained with a variety of DST methods, highlighting methods that both are sparse-to-sparse and learn structured sparsity similar to SRigL — only DSB-16 (2:4 and 1:4 sparsity) is directly comparable in this regard. RigL and SRigL results are from our experiments, other values are obtained from each method’s corresponding paper, U.N.O.

method	training		sparsity				
	method	structured	50%	75%	80%	90%	93.75%
Static*	sparse	no	–	–	70.6 ± 0.06	65.8 ± 0.04	–
SET*	sparse	no	–	–	72.9 ± 0.39	69.6 ± 0.23	–
DeepR [§]	sparse	no	–	–	71.7	70.2	–
DSR	sparse	no	–	–	73.3	71.6	–
Top-KAST [‡]	sparse	no	–	–	74.76	70.42	–
MEST [†]	sparse	no	–	–	75.39	72.58	–
RigL	sparse	no	–	–	74.98	72.81	–
DSB-16	sparse	yes	76.33	74.04	–	–	–
SRigL (Ours)	sparse	yes	76.60	75.55	75.01	72.71	70.56
SNFS (ERK)*	dense	no	–	–	75.2 ± 0.11	73.0 ± 0.04	–
SR-STE	dense	yes	–	76.2	–	–	71.5
dense ResNet-50:			76.7				

* values obtained from [?]

§ values obtained from [?]

† values for the MEST (x0.67+EM) variant, matched to the same number of training FLOPs as RigL

‡ values tabulated for Top-K Always Sparse Training (Top-KAST) correspond to the *backwards sparsity* as Top-KAST uses different sparsities in the forward and backward passes. For 80% sparsity, the value reported is based on using a random regrow criteria, unlike the 90% sparsity value. For more information see Table 1 in [?]

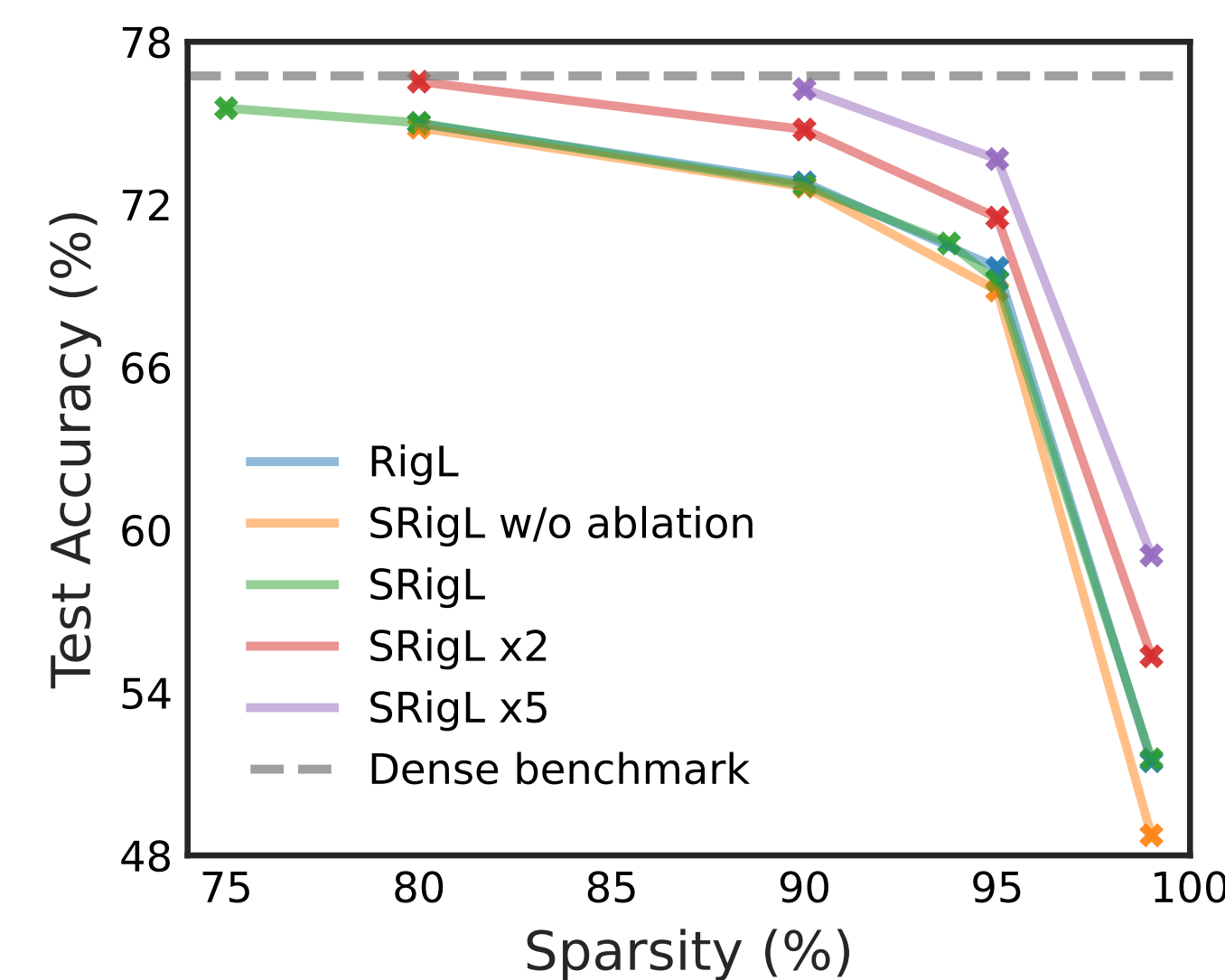


Figure 3. ResNet-50/ImageNet

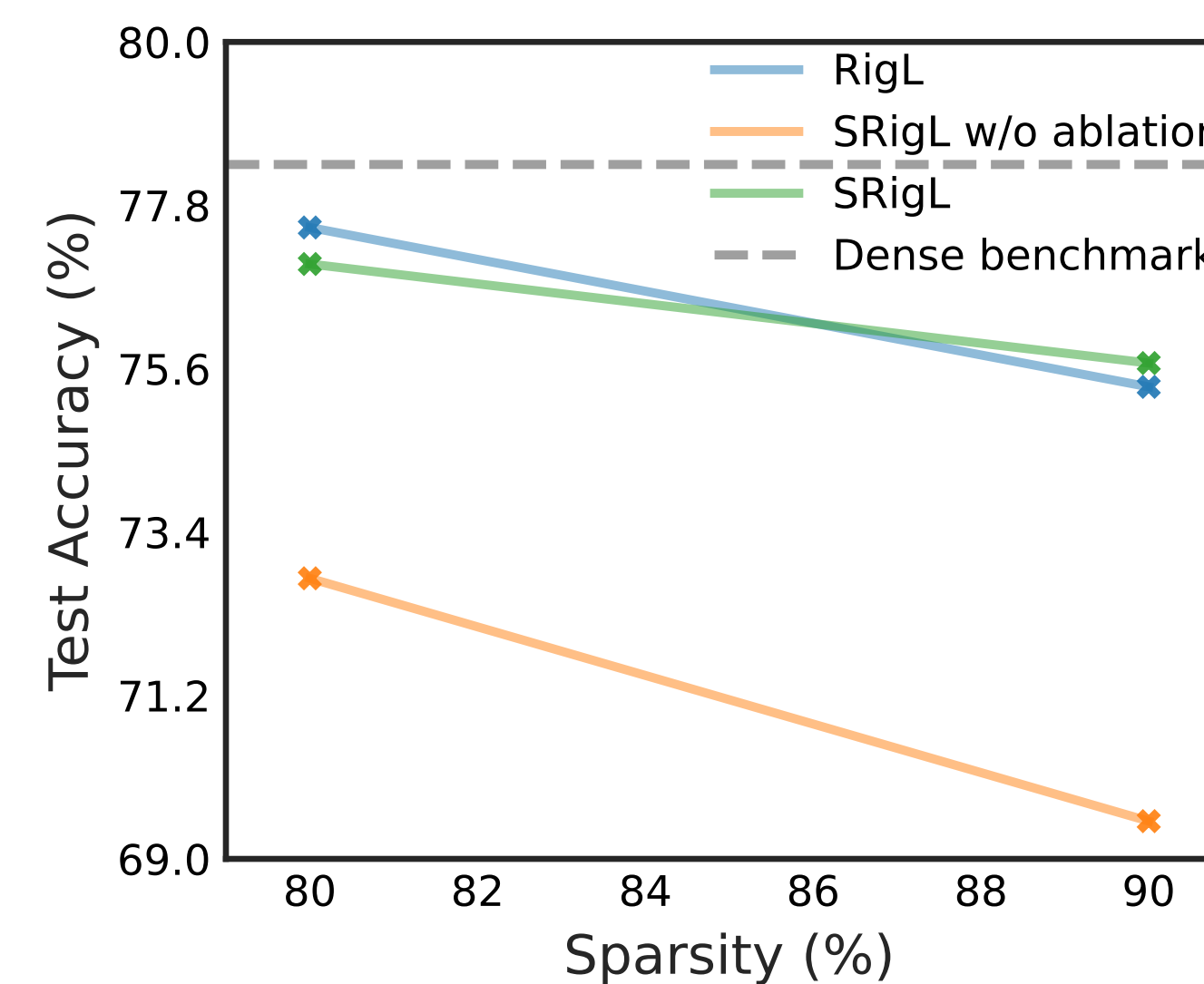


Figure 4. Vision Transformer (ViT-B/16)/ImageNet

FLOPs & Acceleration

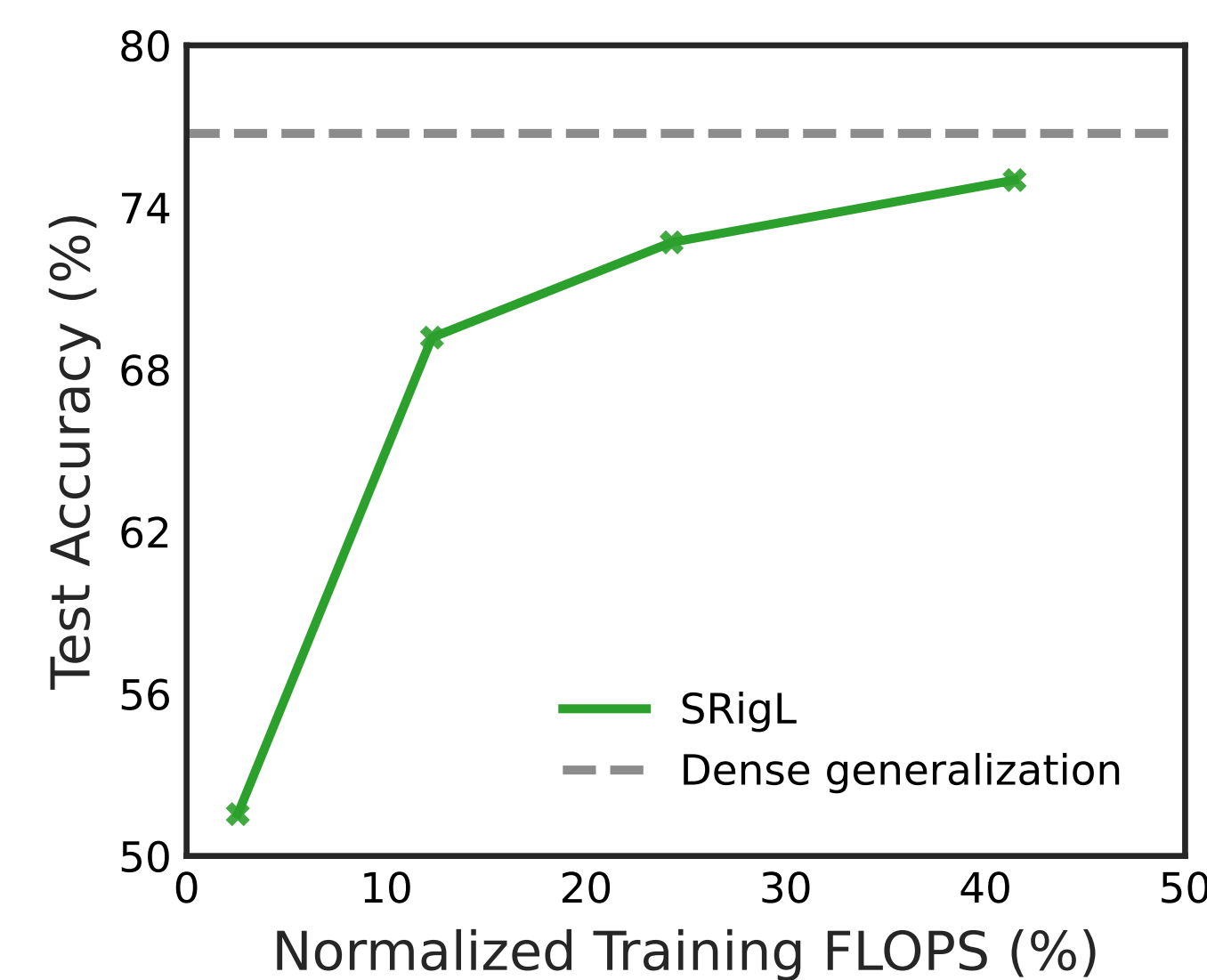


Figure 5. Training FLOPs for SRigL on ResNet-50/ImageNet at a variety of sparsities compared with dense generalization. FLOPs are normalized by dense training FLOPs.

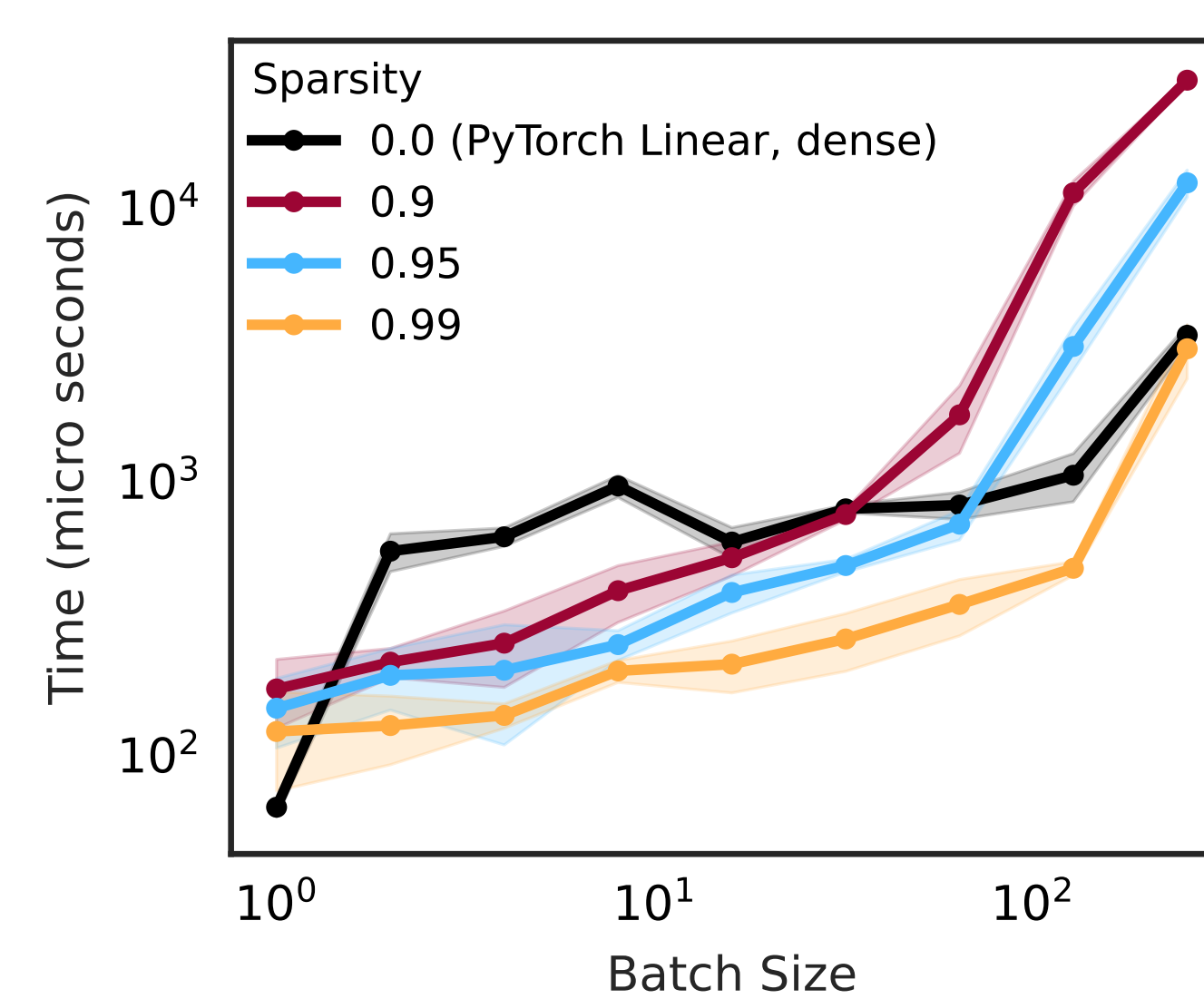


Figure 6. PyTorch timings. Benchmarking a naive (non-optimized) PyTorch implementation of our condensed representation on CPU.

Conclusion

- At high sparsities, RigL ablates large numbers of neurons. Effectively, *RigL reduces the width of the model at high sparsities*.
- Under a strict constant fan-in constraint, neurons can never be ablated. Leading to very sparsely connected neurons at high sparsities.
- We implement a **neuron ablation method**, allowing SRigL to maintain both a constant fan-in constraint *and* to reduce layer width at high sparsities.



Figure 7. Neuron ablation. At sparsity levels over 90%, RigL learns to completely mask (ablate) a large number of neurons within each layer, effectively reducing layer width. Imposing a constant fan-in constraint requires all neurons to have the same number of (non-pruned) incoming weights and therefore inhibits ablation, which results in worse generalization performance than RigL. Allowing SRigL to ablate neurons restores RigL-level performance.

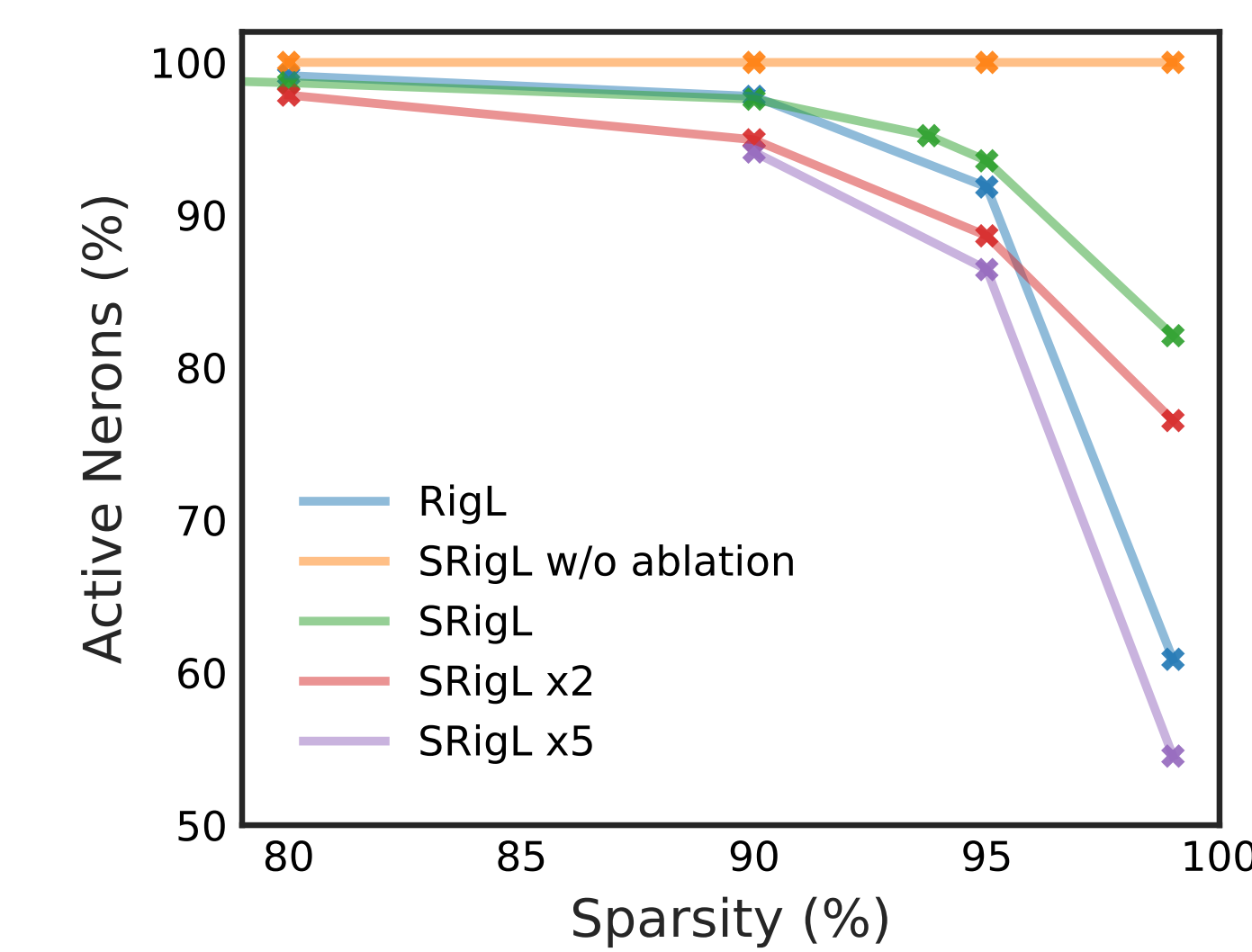


Figure 8. Neuron Ablation

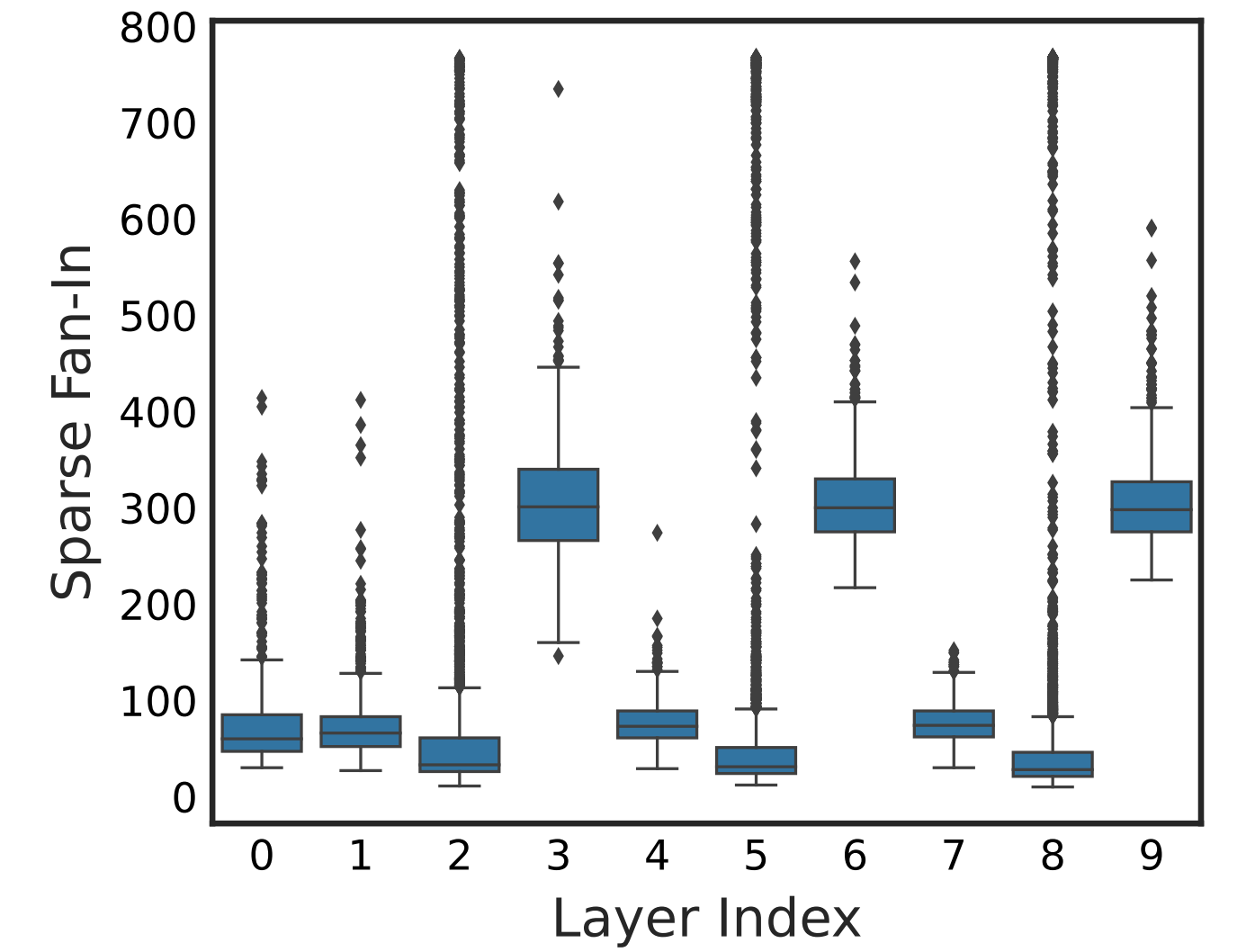


Figure 9. Sparse Fan-In vs. ViT-B/16 layer index at the end of training with RigL at 90% sparsity. Only the first 10 layers are shown for clarity. We find that RigL learns a sparse connectivity with large variance in fan-in between neurons within the same layer with some neurons receiving up to $\times 10$ the number of active connections than the mean for the same layer.

Acknowledgements

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References